

EMG4_justice_trait_data

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packages

```
library(readxl)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.2      v readr      2.1.4
## v forcats    1.0.0      v stringr   1.5.0
## v ggplot2    3.4.2      v tibble    3.2.1
## v lubridate  1.9.2      v tidyr     1.3.0
## v purrr      1.0.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the [8;;http://conflicted.r-lib.org/conflicted-package]8;; to force all conflicts to
o become errors

library(psych)

##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

Questionnaire 2 data

[illegible]

```
## 1 8811 1 2 1 1 4 2 4 4 3 3 2 1
## 2 7735 1 4 1 1 1 1 1 3 2 1 2 1
## 3 7417 1 2 2 2 3 3 3 3 3 3 3 2
## 4 9080 1 3 1 1 2 1 4 4 4 1 2 1
## 5 6012 1 4 4 2 2 1 3 3 3 1 2 5
## 6 1145 2 4 1 1 1 1 1 1 2 2 4 3
## # i 5 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## # `17` <dbl>
```

Select the questions that together measure the construct of Justice

```
justiceQs <- questionnaire_2[, -1]

# Crohnbach's alpha justice questions
# Question 2, 12 and 13 were negatively correlated due to reverse framing. These questions w
ere automatically reversed.
reliability<-psych::alpha(justiceQs, check.keys=TRUE)

## Warning in psych::alpha(justiceQs, check.keys = TRUE): Some items were negatively correla
ted with total scale and were automatically reversed.
## This is indicated by a negative sign for the variable name.

reliability

##
## Reliability analysis
## Call: psych::alpha(x = justiceQs, check.keys = TRUE)
##
## raw_alpha std.alpha G6(smc) average_r S/N ase mean sd median_r
## 0.79 0.8 0.87 0.19 3.9 0.038 2.4 0.48 0.18
##
## 95% confidence boundaries
## lower alpha upper
## Feldt 0.71 0.79 0.86
## Duhachek 0.72 0.79 0.87
##
## Reliability if an item is dropped:
## raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## 1 0.79 0.79 0.87 0.19 3.9 0.038 0.029 0.19
## 2- 0.78 0.78 0.86 0.18 3.6 0.041 0.028 0.17
## 3 0.79 0.79 0.86 0.19 3.7 0.040 0.027 0.19
## 4 0.78 0.78 0.86 0.19 3.6 0.039 0.028 0.17
## 5 0.77 0.77 0.86 0.17 3.4 0.043 0.027 0.17
## 6 0.78 0.79 0.86 0.19 3.7 0.040 0.029 0.18
## 7 0.76 0.77 0.86 0.17 3.4 0.044 0.027 0.16
## 8 0.79 0.79 0.86 0.19 3.7 0.039 0.027 0.19
## 9 0.78 0.78 0.86 0.18 3.6 0.040 0.027 0.18
## 10 0.79 0.79 0.87 0.19 3.8 0.038 0.029 0.19
## 11 0.80 0.80 0.87 0.20 4.0 0.037 0.025 0.19
## 12- 0.79 0.79 0.86 0.19 3.7 0.039 0.025 0.19
## 13- 0.79 0.80 0.86 0.20 3.9 0.038 0.024 0.20
## 14 0.80 0.80 0.87 0.20 3.9 0.037 0.028 0.19
## 15 0.77 0.78 0.85 0.18 3.5 0.042 0.026 0.17
## 16 0.78 0.78 0.85 0.18 3.6 0.041 0.026 0.18
## 17 0.78 0.78 0.86 0.18 3.5 0.040 0.028 0.17
##
## Item statistics
## n raw.r std.r r.cor r.drop mean sd
## 1 60 0.28 0.37 0.30 0.24 1.2 0.38
## 2- 60 0.57 0.55 0.51 0.47 3.5 1.03
## 3 60 0.45 0.47 0.44 0.36 1.7 0.82
## 4 60 0.47 0.50 0.47 0.38 1.8 0.89
## 5 60 0.68 0.67 0.65 0.60 2.6 1.09
## 6 60 0.47 0.49 0.44 0.39 1.6 0.81
## 7 60 0.71 0.67 0.65 0.62 2.6 1.18
## 8 60 0.46 0.46 0.43 0.34 2.8 1.12
```

```
## 9 60 0.53 0.50 0.48 0.44 2.5 0.93
## 10 60 0.38 0.40 0.34 0.28 1.7 0.93
## 11 60 0.27 0.29 0.24 0.15 2.5 0.98
## 12- 60 0.45 0.44 0.42 0.34 4.3 1.03
## 13- 60 0.35 0.34 0.32 0.23 4.0 1.04
## 14 60 0.33 0.34 0.27 0.20 2.2 1.14
## 15 60 0.64 0.61 0.61 0.54 2.3 1.10
## 16 60 0.59 0.55 0.54 0.47 2.5 1.31
## 17 60 0.56 0.59 0.56 0.49 1.6 0.74
##
## Non missing response frequency for each item
## 1 2 3 4 5 miss
## 1 0.83 0.17 0.00 0.00 0.00 0
## 2 0.10 0.53 0.13 0.20 0.03 0
## 3 0.50 0.38 0.07 0.05 0.00 0
## 4 0.45 0.42 0.05 0.08 0.00 0
## 5 0.15 0.40 0.23 0.17 0.05 0
## 6 0.57 0.33 0.05 0.05 0.00 0
## 7 0.12 0.48 0.12 0.20 0.08 0
## 8 0.12 0.32 0.28 0.22 0.07 0
## 9 0.08 0.48 0.25 0.17 0.02 0
## 10 0.50 0.38 0.07 0.02 0.03 0
## 11 0.15 0.42 0.27 0.15 0.02 0
## 12 0.58 0.22 0.12 0.07 0.02 0
## 13 0.38 0.37 0.13 0.10 0.02 0
## 14 0.33 0.35 0.15 0.13 0.03 0
## 15 0.20 0.52 0.10 0.13 0.05 0
## 16 0.23 0.43 0.05 0.18 0.10 0
## 17 0.50 0.38 0.10 0.02 0.00 0
```

The overall Cronbach's alpha is 0.7940737, so we include all questions in our composite Justice score. We do need to reverse the scores for questions 2, 12 and 13.

```
# for final analysis, use question 1, 3:11, 14:17, see "grouping_Justice_traits.xlsx"
justice <- questionnaire_2[, -1] %>%
  select(as.character(c(1, 3:11, 14:17)))

#head(justice)

#Cronbach's alpha selected justice questions
psych::alpha(justice)

# create centered scores so it is clear which ratings are fair/neutral/unfair
# But first reverse the scores for Q 2, 12 and 13
head(questionnaire_2)

## # A tibble: 6 x 18
##   sjCode `1` `2` `3` `4` `5` `6` `7` `8` `9` `10` `11` `12`
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 8811 1 2 1 1 4 2 4 4 3 3 2 1
## 2 7735 1 4 1 1 1 1 1 3 2 1 2 1
## 3 7417 1 2 2 2 3 3 3 3 3 3 3 2
## 4 9080 1 3 1 1 2 1 4 4 4 1 2 1
## 5 6012 1 4 4 2 2 1 3 3 3 1 2 5
## 6 1145 2 4 1 1 1 1 1 1 2 2 4 3
## # i 5 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## # `17` <dbl>

questionnaire_2_centered <- questionnaire_2 %>%
  #select(sjCode, as.character(c(1, 3:11, 14:17))) %>%
  mutate(across(c(3, 13, 14), ~ case_when( #column+1 because column 1 is sjCode
    . == 1 ~ 5,
    . == 2 ~ 4,
    . == 3 ~ 3,
    . == 4 ~ 2,
    . == 5 ~ 1))) %>%
```

```
mutate(across(c(2:18), ~ case_when(
  . == 1 ~ 2,
  . == 2 ~ 1,
  . == 3 ~ 0,
  . == 4 ~ -1,
  . == 5 ~ -2)))
head(questionnaire_2_centered)

## # A tibble: 6 x 18
##   sjCode `1` `2` `3` `4` `5` `6` `7` `8` `9` `10` `11` `12`
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 8811 2 -1 2 2 -1 1 -1 -1 0 0 1 -2
## 2 7735 2 1 2 2 2 2 2 0 1 2 1 -2
## 3 7417 2 -1 1 1 0 0 0 0 0 0 0 -1
## 4 9080 2 0 2 2 1 2 -1 -1 -1 2 1 -2
## 5 6012 2 1 -1 1 1 2 0 0 0 2 1 2
## 6 1145 1 1 2 2 2 2 2 2 1 1 -1 0
## # i 5 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## # `17` <dbl>
```

dataframe with only mean rating scores (trait score) per participant

```
questionnaire_2_means <- questionnaire_2_centered %>%
  pivot_longer(!sjCode, names_to = "Q2", values_to = "J_ratings") %>%
  arrange(sjCode) %>%
  group_by(sjCode) %>%
  summarise(J_score = mean(J_ratings))

head(questionnaire_2_means)

## # A tibble: 6 x 2
##   sjCode J_score
##   <dbl>   <dbl>
## 1 1055 0.235
## 2 1088 0.882
## 3 1124 1.06
## 4 1145 1.18
## 5 1258 1
## 6 1278 1.18

write_csv(questionnaire_2_means, file = paste0(getwd(), "/Qualtrics/justice_means_EMG4_final.csv"))

summary(questionnaire_2_means$J_score)

##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -1.2353 0.2941 0.6471 0.5716 0.9412 1.4706

sd(questionnaire_2_means$J_score)

## [1] 0.4787541
```